

Absolute error



- 1 a 0.5 kg b 0.5 g c \$5
- 2 $\frac{1}{2}$ mL
- 3 a 0.05 mL b 0.5 mm c 0.05 g d 0.5 s
e 0.005 kg f 0.05 m g 0.005 °C
- 4 We do not know what place value the number has been rounded to.
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The limits of accuracy

- 5 a 1 cm b 90 cm
c 0.5 cm d i No ii No iii No iv Yes
- 6 a 10 kg b 5 kg
c Lower bound: 65 kg, upper bound: 75 kg
- 7 a 50 km/h b 5 km/h c 55 km/h
- 8 a i 13.455 km ii 13.445 km b i 6.45 km ii 6.35 km
c i 5.5 m ii 4.5 m d i 14.005 kg ii 13.995 kg
e i 4884.5 m ii 4883.5 m f i \$52.50 ii \$47.50
- 9 19.995 m
- 10 2.05 m

11 a Maximum: 175, minimum: 165



b Maximum: 3050, minimum: 2950

12 a 26.25

b 0.6

13 a 13.85 m/s

b 13.95 m/s

Percentage error

14 a $\pm 0.91\%$ b $\pm 0.96\%$ c $\pm 7.14\%$ d $\pm 1.22\%$
e $\pm 0.29\%$

15 a i 4.7 kg ii 15.5% b i 0.07 m ii 5.3%
c i 7.2 minutes ii 13.8%

16 a 15 891 b 16 109 c 18.9% d 19.2%
e Italy's

Area and volume calculations

17 a 153.9 cm^2 b $132.7 \text{ cm}^2, 176.7 \text{ cm}^2$
c 22.8 cm^2 d 14.8%

18 a Upper bound: 272.6925 m^2 , lower bound: 269.3925 m^2
b Upper bound: 66.2 m, lower bound: 65.8 m

19



- a Lower bound: 9.5 cm, upper bound: 10.5 cm
- b 100 cm^2
- c Lower bound: 90.25 cm^2 , upper bound: 110.25 cm^2
- d 1000 cm^3
- e Lower bound: 857.375 cm^3 , upper bound: 1157.625 cm^3